AVIRAL YADAV

Noida, Uttar Pradesh
+91-7007934595 aviy2402@gmail.com LinkedIn Github Portfolio

EDUCATION

Vellore Institute of Technology Bhopal, Madhya Pradesh
B.Tech - Computer Science and Engineering (Cyber Security and Digital Forensics) | CGPA - 8.01 *2022 - 2026*

SKILLS

Data Analytics & BI: Power BI (Dashboards, DAX, KPI cards), Tableau, Microsoft Excel (Pivot Tables, Formulas), Exploratory Data Analysis (EDA), Data Cleaning & Validation, Data Visualization
SQL & Databases: SQL, MySQL, PostgreSQL
Programming & Development: Python (Pandas, NumPy, Scikit-learn), Java, JavaScript, HTML/CSS, React.js, Node.js, Express.js, Flask, REST APIs
Tools & Version Control: Git, GitHub, BeautifulSoup, Visual Studio Code
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Statistics, Problem Solving

EXPERIENCE

QuantAI [link] May 2025 – August 2025
Data Scientist Intern *Auckland, New Zealand*

- Developed Scrapeazon, a full-stack web scraping platform using Python, Flask, BeautifulSoup, and React to extract and structure product data from paginated Amazon search results.
- Applied web scraping techniques including pagination handling, user-agent rotation, retry logic, and exponential backoff to improve the reliability of data collection.
- Processed and organized scraped data for analysis, gaining hands-on experience in data handling, data structures, cleaning, and visualization workflows.
- Explored data analytics and AI/ML concepts through practical experimentation, including SQL-based analysis, exploratory data analysis, and visualization of structured datasets.

PROJECTS

Scrapeazon — Amazon Product Scraping Platform [link]

- Architected a full-stack application (Python, Flask backend + React frontend) with a REST API layer serving structured product data across 9 attributes per listing.
- BPerformed root-cause analysis on recurring scraping failures (blocked requests, rate limiting) and engineered a retry/backoff + user-agent rotation mechanism, cutting failed-request rate by an estimated 60–70%.
- Built and ran test cases across pagination edge cases and malformed responses to validate data-extraction accuracy.
- Optimized backend processing to bring response time under 10 seconds, directly addressing a performance requirement.

AI Expense Tracker [link]

- Designed and built a full-stack expense-tracking application (React, Node.js, Express.js, PostgreSQL), taking it from requirements analysis through to a deployed, working system.
- Developed 15+ RESTful API endpoints with JWT authentication and input validation; wrote functional test scenarios covering transaction, budget, and auth flows before release.
- Debugged and resolved frontend–backend integration issues (token handling, API error states), improving application stability across 95% of test scenarios.
- Documented API endpoints and setup instructions and deployed via Vercel/Render, owning the full dev lifecycle from design to release

Music Recommender System — Data Pipeline [link]

- Engineered a content-based music recommendation system using Python, Scikit-learn, and Pandas on a dataset of 4,800 songs from 605 artists, generating the top 5 recommendations for each query.
- Applied TF-IDF vectorization and cosine similarity to analyze song metadata and perform similarity-based recommendation and retrieval.
- Connected the Spotify API with an interactive Streamlit application to retrieve real-time album artwork and deliver an end-to-end music search and recommendation experience.

CERTIFICATIONS

- The Bits and Bytes of Computer Networking - Google
- Introduction to Data Science - Infosys Springboard
- Introduction to Modern AI - Cisco Networking Academy